

Rajat Sharma

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EDUCATION

Texas A&M University, College Station, Texas

August 2024-May 2026

Master of Science in Computer Engineering

GPA: 4.0/4.0

- Relevant Coursework: CMOS VLSI Design, Computer architecture, Advanced computer architecture, VLSI design automation, Introduction to hardware design verification, Advanced hardware functional verification, Formal verification.

Bangalore Institute of Technology | Bangalore, India

August 2017-June 2021

B.E. in Electronics and Communications Engineering (Specialization in VLSI)

GPA:8.9/10

- Relevant coursework: VLSI design, Digital system design, Control systems, Digital signal processing, FPGA, Microprocessor.

SKILLS

- **Technical Skills:** RTL, Logic/Circuit Design, Computer architecture, CDC, STA, design verification, SOC design, synthesis.
- **Language's & tools:** Verilog, System Verilog, TCL, Python, C/C++, MATLAB, Perl, System C, UVM, Synopsys VCS, Verdi, Spyglass, PrimeTime, Cadence Virtuoso, Xcelium, vManager, JasperGold, Tempus, Xilinx Vivado/ISE, Lint, Linux, Git, Jira.
- **Technology & Protocol:** DDR controller, RISC/x86, DDR/LPDDR, PCIe, I2C, SPI, UART, AMBA (AXI/AHB/APB).

EXPERIENCE

Design Engineering Intern | Cadence Design Systems, Austin TX

May 2025 – December 2025

- Designed the on-chip Debug system architecture and authored RTL to enable post-silicon validation of memory controller IP.
- Implemented the DDR controller DFI read data path module for robust PHY interface integration.
- Conducted synthesis and performed detailed timing and area analysis, implementing optimizations to meet PPA goals.

Research Assistant | Texas A&M University

August 2024 – April 2025

- Built a 9-stage pipelined ASIC with configurable logic, boosting EMT simulations 25x with FIR/IIR-optimized MAC units.
- Developed RTL with STA-driven optimizations, achieving 100 MHz performance post-synthesis (Xilinx Vivado) as prototype.
- Engineered parallel MAC units to eliminate computational bottlenecks, ensuring cycle-accurate EMT emulation.

Hardware Design Engineer II | Microchip Technology Pvt. Ltd. India

August 2021 - August 2024

- Architected and RTL-designed the DDR SDRAM controller read/write datapath micro-architecture for SoC integration, implementing buffering, pipelining, arbitration, and data alignment logic to optimize bandwidth utilization while meeting strict latency and timing-closure requirements.
- Designed and implemented protocol compliant AXI to AHB bridge logic, including burst decomposition, transaction ordering, and ready/valid handshake control, ensuring correct operation under back pressure and corner case scenarios.
- Performed protocol level verification and debug of DDR datapath and AXI to AHB bridge designs using simulation and assertions, identifying and resolving timing, ordering, and cross-domain issues to ensure full AXI/AHB compliance.

Hardware design Intern | Microchip Technology Pvt. Ltd. India

March 2021- August 2021

- Verified RAID controllers, SAS expanders, and interfaces (I2C, SPI, UART, JTAG, PCIe) during design and DVT phases.
- Devised a Python automation tool for QUAL/DVT testing, reducing test time by 4x and increasing accuracy to 98%.

PROJECTS

The Entangling Instruction Prefetcher

March 2025 – April 2025

- Implemented Entangling Instruction Prefetcher, reducing L1 cache misses by up to 87%.
- Achieved 10.1% IPC improvement using optimized prefetching, validated on wrong-path version of Champ Sim simulator.

The Proximity-aware Cache Coherence Protocol Design

February 2025 – April 2025

- Implemented a proximity-aware MESI protocol to reduce access latency and compared it against MESI and MOESI using a custom C++ simulator on a 16-core mesh; achieved up to 40% lower latency.

Multi-Core Cache Coherency Controller Implementation

January 2025 – March 2025

- Developed Verilog-based MSI cache controller with revamped M/S/I transitions and address decoding for multi-core consistency.
- Managed bus operations (BusRead, BusReadX, BusUpgrade) to resolve conflicts across shared-memory systems.

Design Verification of HTAX Specification using System Verilog & UVM

January 2025 – February 2025

- Developed a modular verification environment for the HTAX crossbar IP with assertions and coverage.
- Ran constrained-random regressions, achieving 100% functional coverage.

AWARDS & CERTIFICATIONS

- System Verilog for Design and Verification, System Verilog Accelerated Verification with UVM, System Verilog Assertions, Python, Deep learning specialization, Cadence VIP, Advanced Synthesis with Genus Stylus.
- Deep Learning Specialization | Coursera (March 2020 – July 2020).
- TensorFlow in Practice Specialization | Coursera (May 2020 – July 2020).